C8 Chemical Analysis – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 1: Purity, Formulations and Chromatography **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Pure substance (chemistry)				Single element or compound
2	Melting point of a pure substance				Sharp, specific value
3	Melting point of an impure substance				Lower and over a range
4	Formulation				Mixture designed as a useful product
5	Examples of formulations				Fuels, paints, medicines, alloys, fertilisers
6	Two phases in chromatography				Stationary and mobile
7	R_f equation				Distance moved by substance ÷ distance moved by solvent
8	Pure substance on a chromatogram				One spot
9	Why the start line is drawn in pencil				Pencil does not dissolve
10	How a substance is identified				Same R_f as a known sample

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C8 Chemical Analysis – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 2: Gas Tests **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Test for hydrogen				Lighted splint: squeaky pop
2	Test for oxygen				Glowing splint relights
3	Test for carbon dioxide				Limewater turns milky
4	Test for chlorine				Damp litmus paper is bleached
5	Limewater				Calcium hydroxide solution
6	Range of R_f values				Between 0 and 1
7	Why the solvent is below the start line				Spots would dissolve into the solvent
8	Stationary phase in paper chromatography				Paper
9	Mobile phase in paper chromatography				Solvent
10	Why a spot stays on the start line				Insoluble in the solvent

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C8 Chemical Analysis – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 3: Flame Tests and Hydroxides **Paper 2** **Triple only**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Flame colour: lithium				Crimson
2	Flame colour: sodium				Yellow
3	Flame colour: potassium				Lilac
4	Flame colour: calcium				Orange-red
5	Flame colour: copper				Green
6	Al^{3+} with NaOH				White precipitate, dissolves in excess
7	Ca^{2+} with NaOH				White precipitate
8	Mg^{2+} with NaOH				White precipitate
9	Cu^{2+} with NaOH				Blue precipitate
10	Fe^{2+} with NaOH				Green precipitate

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

C8 Chemical Analysis – Knowledge Organiser: Self-Test

AQA GCSE Chemistry • Set 4: Negative Ions and Instrumental Methods Paper 2 Triple only

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Fe^{3+} with NaOH				Brown precipitate
2	Test for carbonate ions				Dilute acid; gas turns limewater milky
3	Reagents to test for halide ions				Dilute nitric acid + silver nitrate
4	Chloride with silver nitrate				White precipitate
5	Bromide with silver nitrate				Cream precipitate
6	Iodide with silver nitrate				Yellow precipitate
7	Test for sulfate ions				Dilute HCl + barium chloride: white precipitate
8	Advantages of instrumental methods				Accurate, sensitive, rapid
9	Flame emission spectroscopy identifies ions by				Wavelengths of light emitted (line spectrum)
10	Why flame tests fail for mixtures				Some colours mask others

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.